# HandoffX

Executive research summary | October 28, 2026

A finite-model benchmark and reference library for human intervention under joint uncertainty

## Decision

Retain HandoffX as a comparative research benchmark and decision-support prototype, not a newly invented general theory or a deployment-ready safety system. The completed study asks when continue, retry, human rescue or safe stop minimize explicit costs under imperfect information.

Bayes-adaptive planning already combines hidden states and uncertain parameters (Ross et al., 2011); cost-sensitive deferral and robust dynamic programming are established (Mozannar & Sontag, 2020; Iyengar, 2005). HandoffX makes a small, testable specialization and preserves the conditions under which its conclusions hold.

## What was actually executed

The study specifies persistent hidden task modes, uncertain AI and human capability, fallible terminal human rescue, resource quotas, deadlines and external action restrictions. It compares 11 policies in 38 synthetic scenarios, with exact expectations for all 418 cells and 5,016,000 stored scenario-policy simulation executions. Shared random numbers support paired comparisons; these are not 5,016,000 independent real tasks.

| Evidence | Verified output |
|---|---|
| Mathematics | 9 conditional propositions; 8 counterexamples |
| Software | 60 tests passed; package, wheel and examples executed |
| Reproduction | All 418 numerical rows reproduced; 16,000 scalar trajectories matched |
| Publication | 58-page manuscript; 6 figures; 11 tables |
| Literature | 22 primary metadata records; 6 selected full texts examined |

## Primary finding

The joint-Bayesian reference policy had lower exact cumulative loss than nominal dynamic programming in 6 scenarios, identical loss in 23, and higher loss in 9. Additional uncertainty modeling was therefore not uniformly beneficial. The comparison is conditional on the designed environments, not a population estimate for deployed agents.

## What the findings mean

Persistent-failure modeling mattered more consistently than the chosen cross-capability prior: the Bayes policy improved on the iid-mode ablation in 15 scenarios and tied in 23. The parameter-coupling ablation produced only 1 improvement, 31 ties and 6 degradations. These are differences in exact expected loss, not claims of statistical significance across a random sample of deployments.

Robust planning correctly minimized its own history-rectangular worst-case objective. That objective is more conservative than keeping one unknown model fixed. In the shifted scenario, true robust-policy loss was 13.19452 while its reported envelope was 10.75480: the generating process lay outside the assumed uncertainty set. A robust optimizer cannot repair missing model coverage.

One counterexample shows why a single AI-confidence threshold is insufficient: human competence, remaining time and costs can reverse the decision even when AI confidence is unchanged. Another shows that identical observed AI-only logs can conceal opposite optimal human-intervention decisions when human outcomes are never observed.

## Practical prototype and limits

The handoffx package supplies validated probability and task objects, Bayesian and robust planners, static decisions, guarded action sets, explanations, simulation and result export. It does not learn or calibrate success probabilities. Missing evidence can produce an explicitly unscored stop recommendation only when stopping is authorized externally. It never executes external agent actions.

The final full run took about 28 seconds on the available Linux AMD CPU host, with approximately 176 MiB peak resident memory. Median cold Bayes root planning was about 10.27 ms for these small scenarios. Windows/Intel operation is intended by the portable code but was not executed. No real LLM, human participant, proprietary trace or production integration was evaluated.

## Publication assessment

This is a reproducible research-stage technical report and prototype. It is not externally peer reviewed, not deposited on Zenodo, not a first-of-its-kind theory claim and not a safety certification. Author identity, licensing, independent review, Windows validation and real-agent/human evaluation remain outstanding. The completed evidence does not justify expanding the report to 120-150 pages.

## Selected foundations

Iyengar, G. N. (2005). Robust dynamic programming. Mathematics of Operations Research, 30(2), 257-280. https://doi.org/10.1287/moor.1040.0129

Mozannar, H., & Sontag, D. (2020). Consistent estimators for learning to defer to an expert. Proceedings of Machine Learning Research, 119, 7076-7087.

Ross, S., Pineau, J., Chaib-draa, B., & Kreitmann, P. (2011). A Bayesian approach for learning and planning in partially observable Markov decision processes. Journal of Machine Learning Research, 12, 1729-1770.